



Grid Multiplication 1.

This is how you work out 23×7 using grid multiplication.

	20	3
7	140	21

$$140 + 21 = \underline{161}.$$



- A). What sums are being worked out with these grids ?
You **do not** need to work out the answers !

1). 4	60 8 [] []	2). 5	50 7 [] []	3). 9	20 4 [] []	4). 6	30 9 [] []
5). 8	70 1 [] []	6). 3	80 6 [] []	7). 7	40 2 [] []	8). 9	60 5 [] []

- B). Write down the sum being worked out in each grid.
Copy the grid and then work out the answer.

1). 3	40 2 [] []	2). 6	20 1 [] []	3). 7	30 2 [] []	4). 4	50 4 [] []
5). 4	30 5 [] []	6). 3	60 5 [] []	7). 5	30 7 [] []	8). 6	40 3 [] []
9). 4	50 9 [] []	10). 3	70 6 [] []	11). 6	60 6 [] []	12). 8	70 2 [] []
13). 9	30 6 [] []	14). 8	40 8 [] []	15). 7	30 8 [] []	16). 6	50 8 [] []
17). 9	20 7 [] []	18). 7	90 5 [] []	19). 9	80 3 [] []	20). 7	80 9 [] []

- C). Draw the appropriate grid and then solve the following problems.



- 1). 4×28
6). 7×45
11). 9×23
16). 8×62

- 2). 3×62
7). 4×67
12). 3×57
17). 7×97

- 3). 5×73
8). 8×34
13). 4×93
18). 9×75

- 4). 7×23
9). 3×95
14). 6×84
19). 6×97

- 5). 6×35
10). 5×97
15). 7×68
20). 9×87



You can use grid multiplication to work out much harder sums, such as 36×72 .

	30	6
70	2100	420
2	60	12

$$2100 + 420 + 60 + 12 = \underline{2592}.$$

D). What sums are being worked out with these grids ?
You **do not** need to work out the answers !

1). 40 1	2). 40 7	3). 80 4	4). 90 8																
20	30	60	70																
5	9	3	6																
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5). 40 3	6). 80 6	7). 50 8	8). 60 5																
90	50	20	30																
8	3	8	6																
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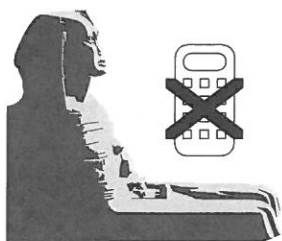
E). Write down the sum being worked out in each grid.
Copy the grid and then work out the answer.

1). 30 6	2). 10 7	3). 40 3	4). 30 4																
20	40	20	50																
3	1	7	3																
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5). 20 6	6). 10 8	7). 30 3	8). 20 4																
30	20	40	60																
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9). 70 4	10). 40 5	11). 20 9	12). 70 8																
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13). 80 3	14). 60 8	15). 70 2	16). 90 4																
20	40	30	50																
6	3	7	1																
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17). 60 2	18). 50 1	19). 30 9	20). 90 5																
70	80	70	20																
6	3	2	9																
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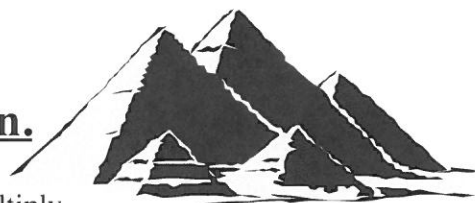
F). Draw the appropriate grid and then solve the following problems.

- | | | | | |
|---------------------|---------------------|---------------------|---------------------|---------------------|
| 1). 34×64 | 2). 57×63 | 3). 38×54 | 4). 93×18 | 5). 41×32 |
| 6). 53×36 | 7). 63×39 | 8). 72×48 | 9). 19×74 | 10). 38×92 |
| 11). 69×48 | 12). 74×38 | 13). 63×85 | 14). 78×69 | 15). 85×59 |
| 16). 97×84 | 17). 76×85 | 18). 49×98 | 19). 96×87 | 20). 79×86 |





Egyptian Multiplication.



Here is a way the Egyptians used to multiply.

An easy example.

To solve 24×33

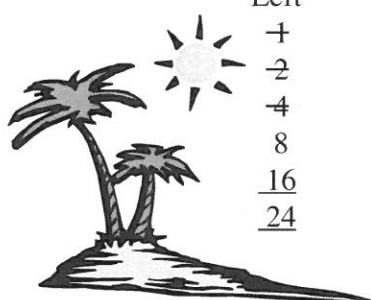
Left	Right
1	33
2	66
4	132
8	264
16	528

Write down a left and right column.

Put 1 in the left column and the bigger number of the sum in the right column.

Keep doubling both columns.....

until the number in the left column is more than half of the smaller number (24).



Left	Right
1	33
2	66
4	132
8	264
16	528
24	<u>792</u>

$24 \times 33 = \underline{792}$.

Now make the smaller number (24) by adding the numbers in the left column. Cross out any we don't need along with those in the right column.

Add up the numbers that haven't been crossed out in the right column and we have our answer.

A Harder Example.

To solve 43×62

Left	Right
1	62
2	124
4	248
8	496
16	992
32	1984

Write down a left and right column.

Put 1 in the left column and the bigger number of the sum in the right column.

Keep doubling both columns.....

until the number in the left column is more than half of the smaller number (43).



Left	Right
1	62
2	124
4	248
8	496
16	992
32	1984
43	<u>2666</u>

$43 \times 62 = \underline{2666}$.

Now make the smaller number (43) by adding the numbers in the left column. Cross out any we don't need along with those in the right column.

Add up the numbers that haven't been crossed out in the right column and we have our answer.

Use this method to solve

1). 12×41

2). 14×32

3). 17×68

4). 24×54

5). 25×35

6). 34×46

7). 41×53

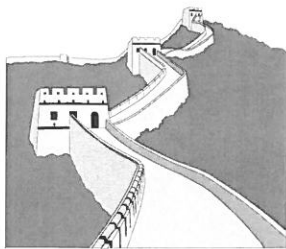
8). 45×61

Hard

9). 29×34

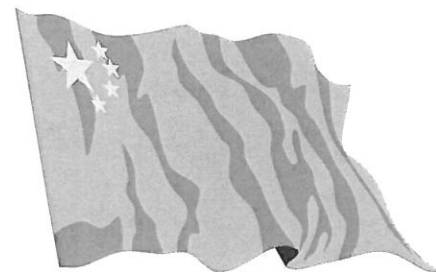
10). 55×72





Chinese Multiplication.

Here is a way the Chinese used to multiply.



An easy example.

Solve 361×46 .

- A). Write the numbers along the top and down the right hand side of a rectangular grid as shown.

3	6	1	
			4
			6

- B). Each pair is multiplied out and the answer put in the relevant box. The tens in the top half and units in the bottom half.

3	6	1	
1 2	2 4	0 4	4
1 8	3 6	0 6	6

- C). The numbers are added up diagonally, starting just with the 6.
The next is $4 + 0 + 6 = 10$. The 0 is put down and the 1 carried to the next diagonal to be added up.

- D). This is now read from the 1. $361 \times 46 = 16606$.

A hard example.

Solve 452×371 .

- A). Write the numbers along the top and down the right hand side of a rectangular grid as shown.

4	5	2	
			3
			7
			1

- B). Each pair is multiplied out and the answer put in the relevant box.

4	5	2	
1 2	1 5	0 6	3
2 8	3 5	1 4	7
0 4	0 5	0 2	1

- C). The numbers are added up diagonally, starting in the bottom right corner.
Remember to carry.

- D). Read off the answer.

$$452 \times 371 = 167692.$$

Try answering these using this method.

- | | | | |
|---------------------|----------------------|-----------------------|---------------------|
| 1). 361×46 | 2). 264×54 | 3). 648×49 | 4). 518×73 |
| 5). 327×98 | 6). 676×57 | 7). 941×83 | 8). 892×74 |
| Hard | 9). 219×134 | 10). 835×472 | |

